Lab # 04

Manipulation of Relational Database Systems using Query Language

Objective:

Student will learn about to do the following:

- Learn the commands to select entries from the table.
- Update and delete entries in table.

Software Tools:

- MySQL Workbench.

Theory:

Relational Database Management System (RDMS): RDBMS is the basis for SQL, and for all modern database systems such as MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access. The data in RDBMS is stored in database objects called tables. A table is a collection of related data entries, and it consists of columns and rows.

Look at the "Customers" table:

SELECT * FROM Customers

Data Manipulation Language:

A **data manipulation language** (**DML**) is a computer programming language used for adding (inserting), deleting, and modifying (updating) data in a database. A DML is often a sublanguage of a broader database language such as SQL, with the DML comprising some of the operators in the language.

DML Commands

Commands	**Working**
UPDATE	Updates the data in the database.
DELETE	Deletes the data from the database.
INSERT INTO	Inserts new data into a database.

QUERIES and COMMANDS for DML

SQL INSERT INTO Statement

The INSERT INTO statement is used to insert new record or row in a table.

SQL INSERT INTO Syntax

It is possible to write the INSERT INTO statement in two forms.

The first form doesn't specify the column names where the data will be inserted, only their values:

```
INSERT INTO table_name
VALUES (value1, value2, value3,...) ;
```

The second form specifies both the column names and the values to be inserted:

```
INSERT INTO table_name (column1, column2, column3,...) VALUES
(value1, value2, value3,...)
```

SQL INSERT INTO Example

We have the following "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

Now we want to insert a new row in the "Persons" table.

We use the following SQL statement:

```
INSERT INTO Persons
VALUES (4,'Nilsen', 'Johan', 'Bakken 2', 'Stavanger')
```

The "Persons" table will now look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger

Insert Data Only in Specified Columns

It is also possible to only add data in specific columns.

The following SQL statement will add a new row, but only add data in the "P_Id", "LastName" and the "FirstName" columns:

```
INSERT INTO Persons (P_Id, LastName, FirstName) VALUES (5,
'Tjessem', 'Jakob')
```

The "Persons" table will now look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob		

The WHERE Clause

The WHERE clause is used to extract only those records that fulfill a specified criterion. The syntax of the command is:

```
SELECT column_name(s)
FROM table_name
WHERE column_name operator value;
```

Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

Now we want to select only the persons living in the city "Sandnes" from the table above.

We use the following SELECT statement:

```
SELECT * FROM Persons
WHERE City='Sandnes' ;
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes

SQL UPDATE Statement

The UPDATE statement is used to update existing records in a table.

SQL UPDATE Syntax

```
UPDATE table_name
SET column1=value, column2=value2,...
WHERE some_column=some_value
```

Note: Notice the WHERE clause in the UPDATE syntax. The WHERE clause specifies which record or records that should be updated. If you omit the WHERE clause, all records will be updated!

SQL UPDATE Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob		

Now we want to update the person "Tjessem, Jakob" in the "Persons" table.

We use the following SQL statement:

```
UPDATE Persons
SET Address='Nissestien 67', City='Sandnes'
WHERE LastName='Tjessem' AND FirstName='Jakob'
```

The "Persons" table will now look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes

2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob	Nissestien 67	Sandnes

SQL UPDATE Warning

Be careful when updating records. If we had omitted the WHERE clause in the example above, like this:

```
UPDATE Persons
SET Address='Nissestien 67', City='Sandnes'
```

The "Persons" table would have looked like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Nissestien 67	Sandnes
2	Svendson	Tove	Nissestien 67	Sandnes
3	Pettersen	Michael	Nissestien 67	Sandnes
4	Nilsen	Johan	Nissestien 67	Sandnes
5	Tjessem	Jakob	Nissestien 67	Sandnes

SQL DELETE Statement

The DELETE statement is used to delete records or rows in a table.

SQL DELETE Syntax

```
DELETE FROM table_name
WHERE some_column=some_value
```

Note: Notice the WHERE clause in the DELETE syntax. The WHERE clause specifies which record or records that should be deleted. If you omit the WHERE clause, all records will be deleted!

SQL DELETE Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob	Nissestien 67	Sandnes

Now we want to delete the person "Tjessem, Jakob" in the "Persons" table.

We use the following SQL statement:

```
DELETE FROM Persons

WHERE LastName='Tjessem' AND FirstName='Jakob'
```

The "Persons" table will now look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger

Delete All Rows

It is possible to delete all rows in a table without deleting the table. This means that the table structure, attributes, and indexes will be intact:

```
DELETE FROM table_name or
DELETE * FROM table_name
```

Note: Be very careful when deleting records. You cannot undo this statement!

SQL SELECT Statement

The SELECT statement is used to select data from a database. The result is stored in a result table, called the result-set.

The syntax used for SELECT query is:

```
SELECT column_name(s) FROM
table_name
```

OR

```
SELECT * FROM table_name
```

Note: SQL is not case sensitive. SELECT is the same as select.

Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

Now we want to select the content of the columns named "LastName" and "FirstName" from the table above.

We use the following SELECT statement:

```
SELECT LastName,FirstName FROM Persons
```

The result-set will look like this:

LastName	FirstName
Hansen	Christ
Svendson	Tove

Pettersen	Michael

SELECT * Example

Now we want to select all the columns from the "Persons" table.

We use the following SELECT statement:

```
SELECT * FROM Persons
```

Tip: The asterisk (*) is a quick way of selecting all columns! The result-set will look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

The SQL SELECT DISTINCT Statement

In a table, some of the columns may contain duplicate values. This is not a problem, however, sometimes you will want to list only the different (distinct) values in a table.

The DISTINCT keyword can be used to return only distinct (different) values.

Its Syntax is:

```
SELECT DISTINCT column_name(s)
FROM table_name
```

Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes

2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

Now we want to select only the distinct values from the column named "City" from the table above.

We use the following SELECT statement:

```
SELECT DISTINCT City FROM Persons
```

The result-set will look like this:

City
Sandnes
Stavanger

The ORDER BY Keyword

The **ORDER BY** keyword is used to sort the result-set by a specified column. The ORDER BY keyword sorts the records in ascending order by default. If you want to sort the records in a descending order, you can use the **DESC** keyword.

Its syntax is:

SELECT column_name(s)

FROM table_name

ORDER BY column_name(s) ASC|DESC

Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

4	Nilsen	Tom	Vingvn 23	Stavanger

Now we want to select all the persons from the table above, however, we want to sort the persons by their last name.

We use the following SELECT statement:

```
SELECT * FROM Persons

ORDER BY LastName
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
4	Nilsen	Tom	Vingvn 23	Stavanger
3	Pettersen	Michael	Storgt 20	Stavanger
2	Svendson	Tove	Borgvn 23	Sandnes

ORDER BY DESC Example

Now we want to select all the persons from the table above, however, we want to sort the persons descending by their last name.

We use the following SELECT statement:.

```
SELECT * FROM Persons

ORDER BY LastName DESC
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger
4	Nilsen	Tom	Vingvn 23	Stavanger
1	Hansen	Christ	Timoteivn 10	Sandnes

Practice all SQL commands described above for any table…!!!

LAB TASK:

A. Create a table Employee having primary key with the following schema:

(Emp_no, E_name, E_address, E_phoneNumber, Dept_no, Dept_name, email_ID, Salary, E_joiningDate, E_serviceYear)

Write SQL query for the following questions:

1. Insert at least 10 rows in the table.
2. Display all the information of the Employee table.
3. Display the Record of each employee who works in the computer department.
4. Update the address of “Emp_no=09” with new address of city Islamabad.
5. Display the name of employee who works in department mechanical.
6. Delete the email ID of employee “James”.
7. Display the complete record of employees with salary of greater than 25,000.

B. Select distinct values for the last three columns from the table below:

Name	Reg_No	Courses	Course_Code	Offered_By
Ali	01	DIP	1001	Mr. A
Basit	02	DBMS	1002	Mr. X
Akram	03	OS	1003	Mr. Y
Asad	04	DBMS	1002	Mr. X
Zeeshan	05	DIP	1001	Mr. A
Muneer	06	OS	1003	Mr. Y
Shafqat	07	NM	1004	Mr. H
Ahsan	08	OS	1003	Mr. Y
Ikram	09	DIP	1001	Mr. A
Hassan	10	DSP	1005	Mr. Z

C. Sort the above table in descending order by their name.

CONCLUSION:

____

____
____